

1 — Departure from CSR: $\tilde{\delta}$ and null tables

1 Theory

Setting

- Window $W = [0, 1]^2$
- $\bar{n}$ = expected number of points in W .
- Point cloud $x = \{x_1, \dots, x_n\} \subset W$; $n(x) = |x|$ denotes realised number of points

Estimator

- Centred L -function of a cloud x :

$$\hat{\phi}_x(r) = \sqrt{\hat{K}_x(r)/\pi} - r, \quad (= 0 \text{ under CSR})$$

where $\hat{K}_x(r)$ is Ripley's edge-corrected K function evaluated at radius r .

- Evaluated on a fixed grid $0 < r_1 < \dots < r_K = 0.25$

Radius set

- For a cloud with $n = n(x)$ points we evaluate $\hat{\phi}_x(r)$ on

$$R(n) = \{r_k : r_k \geq r_{\min}(n)\},$$

where $r_{\min}(n)$ is the smallest radius with at least p expected pairs closer than r under CSR ($\mathbb{P}_{0,n}$):

$$r_{\min}(n) = \min\{r : \mu(r; n) \geq p\}, \quad \mu(r; n) = \frac{1}{2}n(n-1)\pi r^2.$$

- Use $p = 5$. Then

$$r_{\min}(n) = \sqrt{\frac{2p}{\pi n(n-1)}} \approx \frac{1.78}{n}$$

Null distribution

We compute the null distribution at every $n(x)$ on a grid.

- Generate 12 000 CSR clouds from $\mathbb{P}_{0,n}$
- Estimate null moments for $r \in R(n)$ using half of the CSR clouds:

$$m_0(r; n) = \mathbb{E}_{0,n}[\hat{\phi}_X(r)], \quad s_0(r; n) = \text{sd}_{0,n}[\hat{\phi}_X(r)]$$

- Using remaining half of CSR clouds, compute maximum deviation of each cloud x :

$$M_n(x) = \max_{r \in R(n)} \frac{|\hat{\phi}_x(r) - m_0(r; n)|}{s_0(r; n)}$$

- Determine critical value: $c_{95}(n) = 95\%$ quantile of M_n over these clouds.
- Repeated for every $n(x)$ on grid, then smooth m_0 , s_0 , c_{95} across $n(x)$ (cubic splines in $\log n$)

Studentised global envelope (Myllymäki et al. 2017)

- Reject CSR at 5% iff $M_n(x) > c_{95}(n)$, $n = |x|$
- We define the "distance from CSR" $\tilde{\delta}$:

$$\tilde{\delta}(\theta, \bar{n}) = \frac{1}{c_{95}(\bar{n})} \max_{r \in R(\bar{n})} \frac{|L_\theta(r) - r|}{s_0(r; \bar{n})}$$

- $\tilde{\delta} = 0$ at CSR. $\tilde{\delta} = 1$ is on the rejection boundary
- This property is known before simulation, so is usable as a sweep coordinate
- At fixed shape, $\tilde{\delta}$ grows with $\bar{n}$ (s_0 shrinks)